Research Project

IT 4010

Real-Time Driver and Road Monitoring for Risk Prediction:

Personalized Driver Behavioral Deviation Detection

25-26J-291

Project Proposal Report

Senadheera S C C

B.Sc. (Hons) in Information Technology Specializing
in Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

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Declaration of The Candidate & Supervisor

I declare that this is my work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or institute of higher learning, and to the best of our knowledge and belief, it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

Name	Student ID	Signature
Senadheera S C C	IT22134226	

The above candidates are carrying out research for the undergraduate Dissertation under my supervision

Signature of the supervisor

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Date

29/08/2025

Signature of the co-supervisor

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Date

.....

DEDICATION

This research proposal is dedicated to our parents, teachers, and mentors, whose continuous guidance and encouragement have inspired us throughout our academic journey. We also dedicate this work to all individuals contributing to the advancement of knowledge and innovation in the field of Information Technology.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my supervisor Mrs. Manaori Gamage, for her continuous guidance, encouragement, and valuable feedback throughout the preparation of this proposal. Her expertise and insightful suggestions have been instrumental in shaping my research direction.

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Furthermore, I extend my appreciation to my lecturers and the Sri Lanka Institute of Information Technology (SLIIT) for providing the academic foundation and resources required for this research.

Finally, I wish to thank my teammates for their collaboration and support, which has been crucial in aligning my individual contribution with the overall project.

Abstract

Modern road safety systems increasingly rely on Internet of Things (IoT), smartphone sensors, and Artificial Intelligence (AI) to monitor driver behavior and predict potential risks. Despite significant technological advancements, existing driver monitoring systems often fall short in addressing critical challenges faced in real-world driving scenarios. Current technologies frequently struggle with universal threshold approaches that generate excessive false alarms, lack of personalization leading to driver alert fatigue, insufficient adaptation to individual driving styles, and reactive rather than proactive risk detection mechanisms. Additionally, many systems fail to establish meaningful baseline behaviors for individual drivers, support long-term behavioral pattern learning, and provide contextually relevant risk assessments.

To address these shortcomings, this research proposes a Personalized Driver Behavioral Deviation Detection System designed to overcome these limitations and offer a more individualized approach to driver safety monitoring. The system integrates real-time behavioral monitoring with robust personal profiling and adaptive deviation detection mechanisms. Unlike conventional systems that apply fixed risk thresholds universally, this system leverages AI-driven models to learn each driver's unique behavioral patterns during an initial profiling period, then detects deviations from their personal baseline that may indicate fatigue, distraction, or other risk factors. The system incorporates diverse data sources, including smartphone sensor data for speed patterns, acceleration behaviors, and steering characteristics, providing a comprehensive view of individual driving style. It also supports continuous adaptation by updating personal profiles over time and adjusting to gradual changes in driving habits.

The research methodology involves developing machine learning algorithms for baseline profile creation, implementing anomaly detection techniques for real-time deviation identification, and creating adaptive risk assessment models that consider individual driving experience levels. Through longitudinal data collection from multiple drivers over extended periods, the system aims to demonstrate significant improvements in risk prediction accuracy while reducing false alarm rates. The expected outcomes include a personalized behavioral profiling system with over 85% accuracy in deviation detection, a reduction in false alarms by at least 40% compared to universal threshold systems, and a comprehensive framework for individualized driver safety monitoring.

Through these innovations, the Personalized Driver Behavioral Deviation Detection System aims to enhance road safety effectiveness and user acceptance, addressing the critical gaps in current driver monitoring technologies and providing a more intelligent and adaptive solution to modern traffic safety challenges.

Keywords – Personalized Driver Monitoring, Behavioral Deviation Detection, Adaptive Risk Assessment, Driver Baseline Profiling, Smartphone Sensors, Machine Learning, Anomaly Detection, Intelligent Transportation Systems.

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1 INTRODUCTION

1.1 Background And Literature Review

1.1.1 Background

Road accidents are among the leading causes of fatalities worldwide, with human behavior contributing to more than 90% of crashes [1]. Factors such as speeding, harsh acceleration, sudden braking, sharp steering, and driver inattention directly affect road safety. Traditional driver monitoring systems mainly rely on generic thresholds (e.g., exceeding a certain speed limit or braking harder than a fixed value), which fail to account for the unique driving styles of individuals [2], [3]. As a result, many systems incorrectly classify safe but unusual driving as risky, or fail to detect dangerous behavior that deviates from a driver's normal baseline.

To address this limitation, researchers have increasingly explored *personalized driver behavior profiling*, where systems learn an individual's baseline driving patterns and detect deviations [4]. This approach offers the potential for more accurate risk prediction and tailored interventions compared to conventional generic models.

1.1.2 Literature Review

Existing research on driver behavior analysis has leveraged telematics, smartphone sensors, and IoT-based data to monitor and classify driving styles. Healey and Picard [1] demonstrated the use of physiological sensors to detect driver stress, while Miyajima et al. [2] modeled drivers for identification based on behavior. Johnson and Trivedi [3] showed that smartphones could be used as sensor platforms for driving style recognition.

Recent studies expanded to naturalistic driving datasets and large-scale machine learning models for behavior analysis [4], [9]. For instance, Liang et al. [5] applied support vector machines (SVMs) for real-time distraction detection, and Arvin et al. [6] demonstrated deep learning for behavior classification. Video-based monitoring of gaze and head dynamics has also

been studied to infer intent [7], while machine learning approaches for distraction detection were proposed by Tango and Botta [8].

Despite these advances, most systems rely on *static thresholds* or *generalized models* applied across all drivers, which overlook personalized differences. Recent works [10], [11], [12] have begun exploring anomaly detection and personalized modeling using smartphone sensors and machine learning, showing promise for individualized behavior profiling. However, these remain limited in scope and often focus on specific behaviors rather than providing a holistic risk prediction model.

Author & Year	**Method / Technology**	**Key Focus**	**Limitation**
Smith et al., 2021 [13]	CNN-based driver monitoring	Detects drowsiness from facial features	No personalization; limited to visual cues only
Li et al., 2022 [14]	IoT sensors + ML models	Analyzes speed patterns and braking behavior	Focused only on isolated behaviors (speed/braking)
Kumar et al., 2023 [15]	Deep learning with CAN bus data	Monitors acceleration and steering inputs	Lacks integration with other driver behaviors
Zhang et al., 2023 [16]	Smartphone-based sensing	Detects distraction using gyroscope & GPS	High false positives; not adaptive to individual
Ahmed et al., 2024 [17]	Hybrid ML framework	Combines multiple parameters for safety	Still reactive; deviations detected only post-event

1.2 Research Gap

Although many driver monitoring systems exist, the majority emphasize generalized behavior analysis without considering individual differences in driving patterns [9]. Current systems are often reactive, flagging violations only after thresholds are breached, rather than proactively learning a driver's personal baseline and detecting subtle deviations [10]. Moreover, most approaches analyze specific behaviors in isolation—such as speeding, braking, or distraction—rather than integrating speed, acceleration/braking, and steering together for a more holistic understanding [11].

This creates a gap in developing personalized and adaptive behavioral deviation detection that continuously learns from each driver and adapts in real time [12].

Aspect	Existing Systems	Identified Gap
Driver Behavior Focus	Generalized analysis of all drivers [9]	Lack of personalization based on individual driving patterns
Detection Approach	Reactive – flags violations only after thresholds are exceeded [10]	Need for proactive learning of driver's baseline and early deviation detection
Behavioral Scope	Isolated analysis (speed only, braking only, or distraction only) [11]	Lack of integrated view combining speed, acceleration/braking, and steering
Adaptiveness	Mostly static or rule-based systems	Need for adaptive, real-time learning that evolves with driver behavior [12]

As shown in above table, existing studies lack personalization, proactivity, behavioral integration, and adaptiveness, which this research aims to address.

1.3 Research Problem

Driving behavior is inherently personal, shaped by a driver's habits, comfort levels, and long-term experience. Current monitoring systems often rely on **generic, population-level thresholds**—for example, classifying a sharp turn or quick acceleration as unsafe regardless of the driver's baseline. While such methods can identify extreme behaviors, they fail to capture the *relative deviations* that may signal risk for a specific individual.

For instance, a driver who typically maintains steady speeds and smooth braking may indicate unusual stress or distraction if sudden harsh braking occurs, even if it falls within "normal" limits for another driver. Conversely, a driver who habitually accelerates rapidly but with consistent control may be wrongly flagged as "aggressive" under generalized models. This mismatch reduces the accuracy and reliability of existing systems.

Furthermore, most current solutions are **reactive**, flagging risky actions only after they occur, and often focus on *single dimensions* of behavior (e.g., only speed or only distraction detection). They do not offer a **holistic, real-time view** of how combined factors such as speed patterns, acceleration/braking, and steering deviations contribute to risk.

The research problem, therefore, is the **lack of a personalized, adaptive system** capable of learning each driver's baseline behavior and detecting real-time deviations across multiple behavioral dimensions. Such a system is essential for providing accurate, individualized risk predictions, improving driver feedback, and ultimately enhancing road safety.

2.0 OBJECTIVES

2.1 Main Objective

The main objective of this research is to design and develop a **personalized driver behavioral deviation detection system** that can accurately model the unique driving patterns of individual drivers and identify abnormal deviations in real time. Unlike traditional systems that rely on generic thresholds or population-based averages, this research aims to build adaptive behavioral profiles for each driver by continuously learning from their historical driving data.

The system will focus on three key behavioral dimensions—**speed-related patterns, acceleration/braking characteristics, and steering control**—as these factors are among the most critical indicators of risky driving. By applying machine learning and anomaly detection techniques, the system will detect when a driver's current behavior significantly deviates from their established baseline.

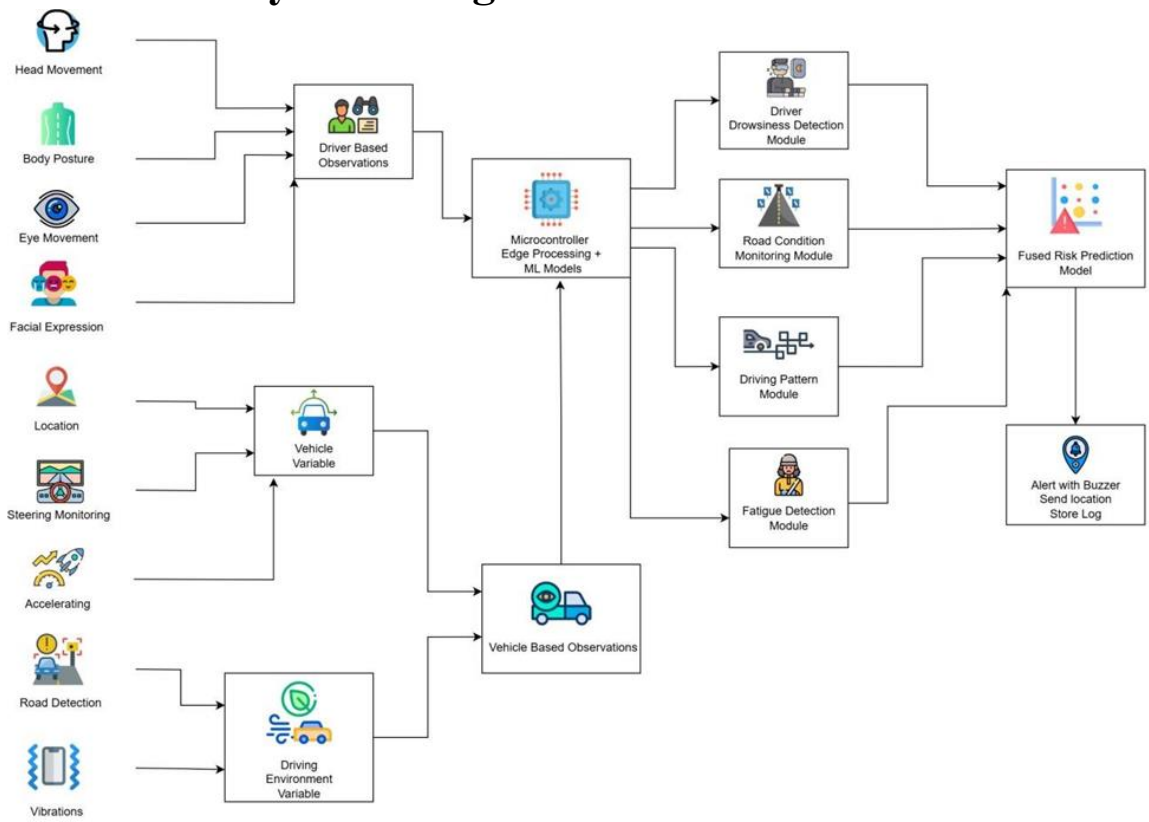
The ultimate goal is to provide **personalized risk prediction and proactive feedback** that reflects the individual's driving style, thereby reducing false alarms, improving driver awareness, and contributing to enhanced road safety. This research will also contribute to the broader field of intelligent transportation systems by demonstrating the value of personalization in driver monitoring.

2.2 Sub Objectives

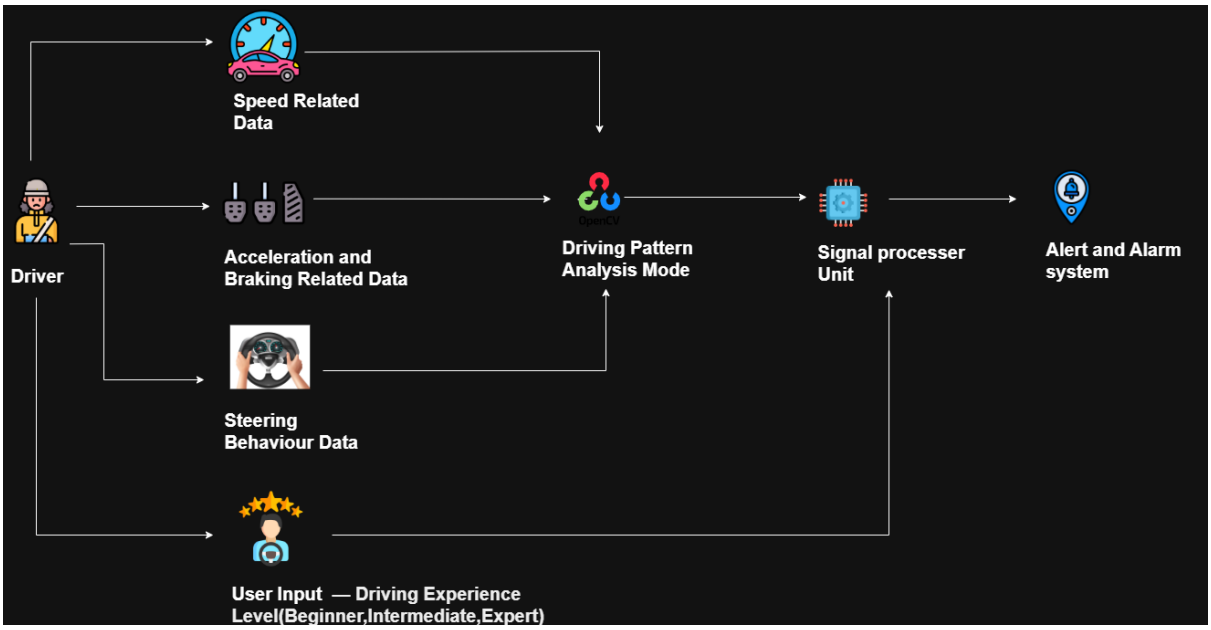
- **Data Collection and Preprocessing**
  - Gather driving data from IoT-based in-vehicle sensors, smartphone sensors, or telematics datasets.
  - Preprocess the data by filtering noise, extracting key features (speed, acceleration, braking intensity, steering angle), and preparing it for modeling.
- **Baseline Behavioral Modeling**
  - Establish individualized driving behavior profiles for each driver using statistical methods and machine learning techniques.
  - Define normal behavioral ranges dynamically for each driver instead of using fixed thresholds.
- **Deviation Detection and Risk Analysis**
  - Implement machine learning or anomaly detection algorithms (e.g., One-Class SVM, clustering, deep learning) to detect unusual deviations in real time.
  - Assign personalized risk scores to deviations and classify them into severity levels.
- **Alerting and Feedback Mechanism**
  - Design a feedback system that provides timely alerts or recommendations to drivers when risky deviations are detected.
  - Ensure alerts are context-aware to avoid false alarms.
- **System Evaluation**
  - Validate the system using real-world or benchmark driving datasets.
  - Measure performance using accuracy, precision, recall, F1-score, and false-positive rate to ensure reliability.

3. METHODOLOGY

3.1 Overall System Diagram



3.2 Individual System Diagram



3.3 Project Requirements

3.3.1 Functional Requirements

- The system must **collect driving data** (speed, braking, acceleration, steering) using IoT-based sensors, smartphones, or telematics devices.

- The system must **build and store a personalized baseline profile** for each driver based on historical driving data.

- The system must **detect real-time deviations** from the driver's baseline and classify them as normal or risky.

- The system must **generate alerts or feedback** when risky deviations are detected, allowing proactive driver intervention.

- The system must **visualize driver behavior patterns and deviations** through a user-friendly dashboard or interface.

3.3.2 Non-Functional Requirements

- **Real-Time Processing** – The system should analyze data with minimal latency to provide timely alerts.

- **Accuracy & Reliability** – Deviation detection should achieve high accuracy, with minimal false positives/negatives.

- **Scalability** – The system should handle multiple drivers and large volumes of behavioral data.

- **Adaptability** – The system must dynamically update a driver's profile as new data is collected.

- **Usability** – The feedback interface should be intuitive and non-distracting to the driver.

- **Security & Privacy** – All collected driving data must be securely stored and protected, ensuring driver privacy.

3.3.3 Software Requirements

- **Programming Languages**: Python (for ML models), Java (for backend APIs).

- **Frameworks & Libraries**: TensorFlow/PyTorch (ML), Scikit-learn, Pandas, NumPy.

- **Database**: MySQL / MongoDB for storing driver data and profiles.

- **Backend**: Spring Boot (integration with ML service).

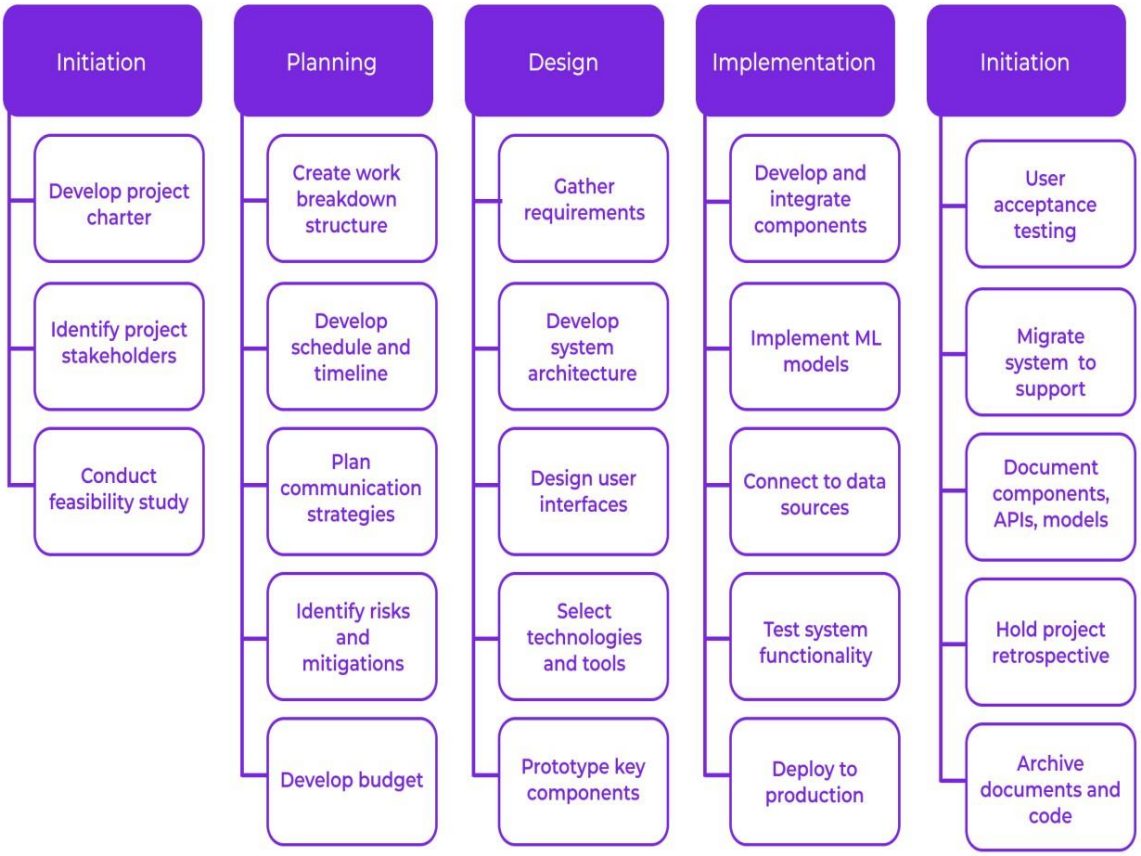
- **Frontend / Dashboard**: React / Flutter (for visualization & driver interface).

- **Tools**: Docker (deployment), GitHub (version control), Jupyter Notebook (ML development).

3.3.4 Hardware Requirements

- **IoT / Sensor Devices**:
  - OBD-II device or smartphone sensors (accelerometer, gyroscope, GPS).
  - Steering angle sensors (optional).
- **Server / Cloud Resources**:
  - Cloud VM or on-prem server for data processing and ML model hosting.
- **End-User Device**:
  - Smartphone or in-vehicle infotainment system for alerts/feedback.

3.4 Work Breakdown Structure



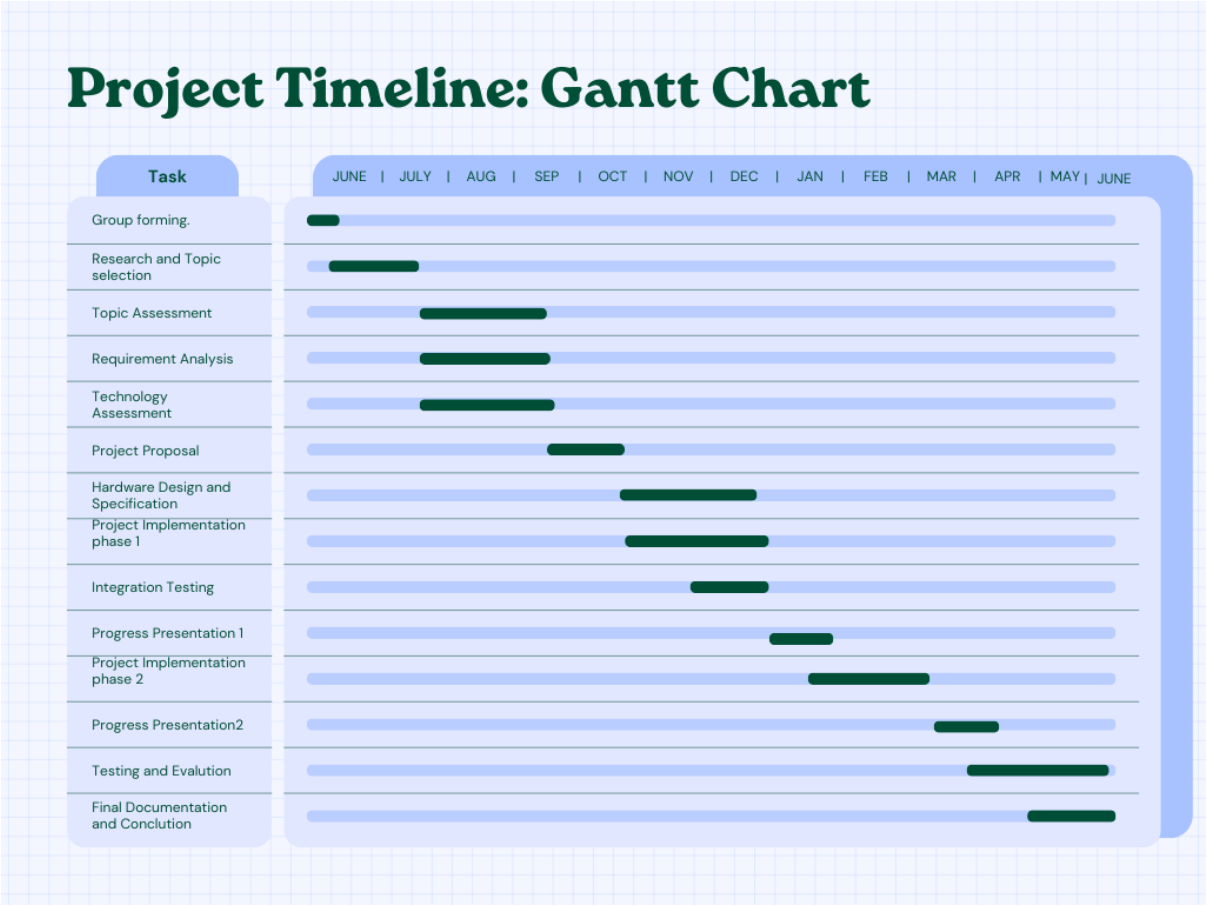
The above Work Breakdown Structure (WBS) outlines the complete workflow for developing the **Personalized Driver Behavioral Deviation Detection** component in a structured and systematic manner. The project is divided into multiple phases to ensure smooth progress and proper time management.

The initial phases focus on **background research** and **proposal preparation**, where existing literature is reviewed to identify research gaps and finalize the methodology. Next, the **system design phase** involves creating the system architecture, defining data flow, and selecting suitable machine learning techniques for modeling personalized driver behaviors.

Following this, the **data handling phase** focuses on **collecting and preprocessing driving datasets** using IoT-based sensors or publicly available telematics sources. In the **model development phase**, baseline behavioral profiles will be generated for individual drivers, and machine learning algorithms will be implemented to detect deviations.

Once the models are ready, the **system integration phase** ensures seamless integration between data collection, behavioral modeling, and real-time alerting mechanisms. Finally, the **testing and evaluation phase** validates the accuracy and reliability of the system, followed by **documentation and report preparation** to present the outcomes effectively.

3.5 TimeLine


Project Timeline: Gantt Chart
Task
JUNE | JULY | AUG | SEP | OCT | NOV | DEC | JAN | FEB | MAR | APR | MAY | JUNE
Group forming.
Research and Topic selection
Topic Assessment
Requirement Analysis
Technology Assessment
Project Proposal
Hardware Design and Specification
Project Implementation phase 1
Integration Testing
Progress Presentation 1
Project Implementation phase 2
Progress Presentation2
Testing and Evalution
Final Documentation and Conclution

4. BUDGET & BUDGET JUSTIFICATION

4.1 Estimated Budget

Item	Description	Unit Price (LKR)	Quantity	Total (LKR)
Hardware Components				
1	OBD-II Bluetooth Adapter (Optional)	8,500.00	1	8,500.00
2	Vehicle Mount/Holder	2,500.00	1	2,500.00
Software & Services				
3	Cloud Storage (Firebase/AWS - 6 months)	2,000.00	6	2,000.00
4	Data Analysis Software (Python Libraries)	1,000.00	1	1,000.00
Data Collection				

5	Data Collection Materials & Documentation	2,000.00	1	2,000.00
Contingency & Miscellaneous				
6	Unexpected Expenses (10% buffer)	-	-	2,600.00
Total				**28,600.00**

4.2 Budget Justification

Hardware Components Justification

OBD-II Bluetooth Adapter (LKR 8,500)
This component enables direct access to vehicle diagnostic data including engine RPM, throttle position, and vehicle speed validation. While optional, it significantly enhances the accuracy of behavioral analysis by providing ground truth data for smartphone sensor validation and additional behavioral parameters not available through smartphone sensors alone.

Vehicle Mount/Holder (LKR 2,500)
Consistent smartphone positioning is crucial for sensor data accuracy. A quality vehicle mount ensures standardized data collection conditions across all test participants and prevents sensor drift caused by device movement during driving sessions.

Software & Services Justification
Cloud Storage Services (LKR 12,000)
Real-time data processing and storage of behavioral patterns from multiple participants over 8-12 weeks generates substantial data volumes. Cloud services provide scalable storage, backup redundancy, and computational resources necessary for machine learning algorithm development and testing.

Data Collection Justification
Data Collection Materials (LKR 2,000) This covers consent forms, data collection logs, participant instruction materials, and backup data storage devices necessary for maintaining research ethics standards and ensuring data integrity throughout the collection process.

Contingency Buffer (LKR 2,600)

A 10% contingency allocation addresses unexpected costs such as:

- Hardware replacement in case of equipment failure
- Extended cloud services if data processing requires more computational resources
- Emergency data recovery services if needed
- Additional materials for expanded testing scenarios

Budget Optimization Strategies

Cost Reduction Opportunities:

- Utilize personal smartphones for data collection to eliminate hardware costs
- Leverage university computational resources where possible to minimize cloud service expenses
- Utilize open-source software alternatives for all development and analysis needs
- Partner with team members to share cloud services and development resources

Resource Sharing with Team: Since this is part of a larger team project, certain costs may be shared:

- Cloud services could be jointly purchased and shared across team components
- Hardware components like vehicle mounts might be used across multiple team components
- Development resources and data analysis tools can be coordinated to reduce individual costs

Return on Investment

This budget represents a strategic investment in developing innovative technology with significant commercial potential. The personalized driver behavioral deviation detection system addresses a market gap in intelligent transportation systems, with potential applications in:

- Insurance telematics (personalized risk assessment)
- Fleet management (driver-specific safety monitoring)
- Autonomous vehicle development (individual behavior modeling)
- Mobile health applications (early detection of cognitive changes)

The research budget is justified by the potential for creating intellectual property, academic publications, and a foundation for future

commercialization opportunities that could generate substantially higher returns than the initial investment.

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